[scale=0.85] [font=, black!80] at (11.5,12.8) **Why the Optimiser Chose $n = 12$ Despite Beam-Only Sc**

[shift=(0,0)] [red!3, rounded corners=6pt] (-0.5,-2.0) rectangle (9.0,11.2); [font=, red!60!black] at (4.

//ıin 3/1.00/1.00/0, 6/1.52/1.52/1, 8/1.77/1.77/2, 10/1.99/1.99/3, 12/2.19/2.19/4 * 3.0 1.2 + ı* 1.

[font=, red!70!black, fill=white, draw=red!30, rounded corners=3pt, inner sep=3pt] at (2.8,8.0) n=3

[-¿, line width=4pt, ¿=stealth, magenta!60] (9.4,5.5) – (11.0,5.5); [font=, magenta!60, align=center]

system

effects;

[shift=(11.5,0)] [blue!2, rounded corners=6pt] (-0.5,-2.0) rectangle (9.5,11.2); [font=, blue!60!black] a

[-¿, thick] (0.8,1.5) – (0.8,9.0) node[above, font=] Total mass (kg); [-¿, thick] (0.8,1.5) – (9.2,1.5) nod

; /in 3/2.2,6/3.8,8/5.8,10/7.4,12/9.0 [font=] at (,1.1) ;

[blue!60, line width=3pt] plot coordinates (2.2,7.0) (3.8,5.2) (5.8,3.8) (7.4,2.8) (9.0,2.0) ; [blue!50] (2

[font=, blue!70!black, align=left, fill=white, draw=blue!20, rounded corners=3pt, inner sep=4pt] at

1. Thinner beams at higher n
2. Smaller knuckles (coupled to D_o)
3. Tether ↑ modest, rotor ↑ modest
→ system optimum at n=12;

[black!3, rounded corners=6pt] (-0.5,-2.5) rectangle (21.0,-1.5); [font=, black!70, align=center] at (10